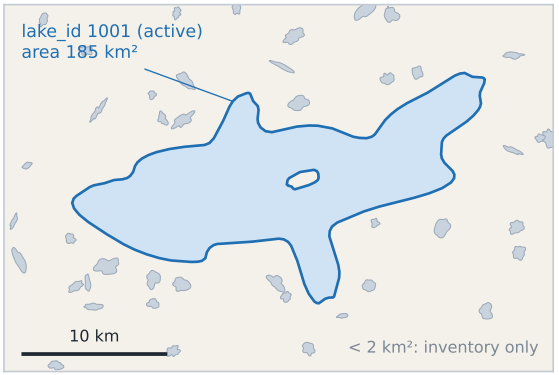
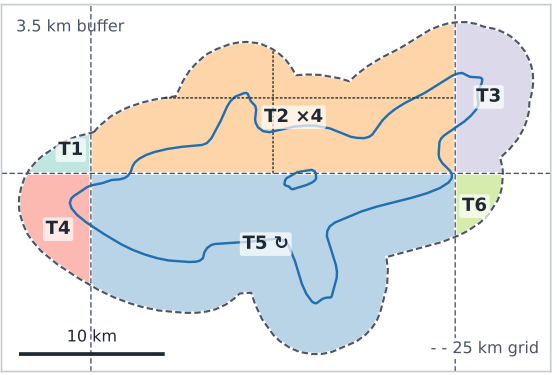
1 Lake synchronisation



HydroLAKES polygons → PostGIS lakes table; active cohort: area ≥ 2 km² (smaller lakes kept for the map only)

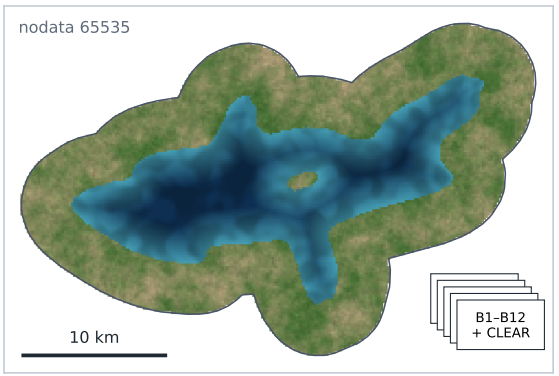
2 Imagery acquisition



×4: memory limit → 4 parts, stitched ↻: HTTP 429 → next account

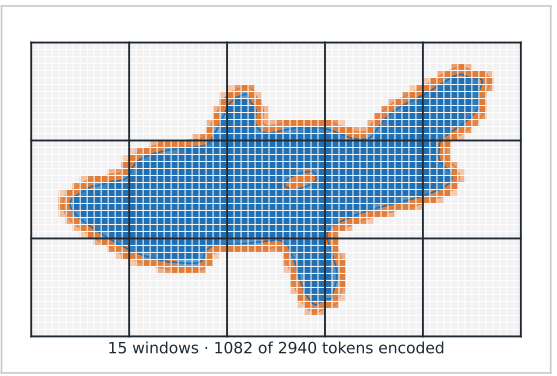
3.5 km buffer, 25 km tiles on a fixed global 30 m grid; S2 L2A, scene cloud ≤ 20 %, Cloud Score+ ≥ 0.6, annual median

3 Tile merging



Merged once all tiles of the lake-year are complete → lake_<id>.tif, 12-band uint16 (reflectance × 10⁴)

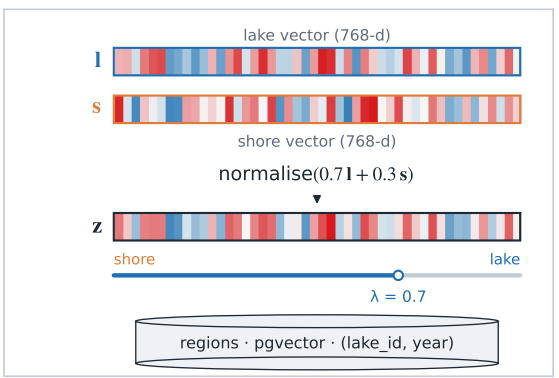
4 Embedding generation



lake token ($a_t > 0$) shore token ($b_t > 0$) not encoded

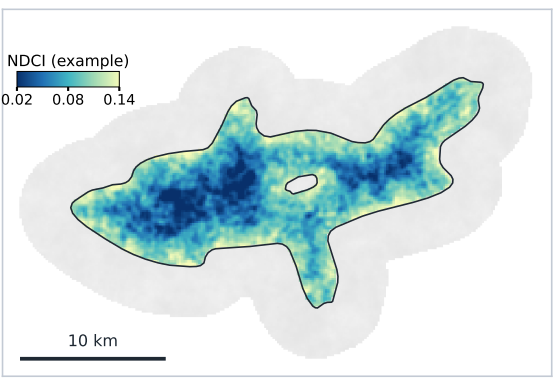
224 × 224 px windows (6.72 km) of 480 m tokens; only tokens touching the lake or 500 m shore ring go to Prithvi-EO-1.0

5 Embedding re-blending



z = normalise(λ·l + (1 − λ)·s), λ = 0.7, stored with l and s in regions (pgvector); a new λ re-blends without inference

6 Lake-feature generation



shore / buffer pixels: excluded lake pixels: statistics

From the merged raster (3), lake pixels only: 11 bands × 5 statistics, 14 indices, GLCM on B8 → lake_features

Stages run in this order as background API jobs; each records per-tile / per-lake-year status in PostgreSQL and skips work that is already complete and current (resumable). Illustrative synthetic lake, drawn to scale.